PERFECT AND EXTREME BINARY HUMBERT FORMS OVER REAL QUADRATIC FIELDS IN THE MULTIPLICATIVE NORMALISATION: AN ENUMERATION FOR 157 FIELDS, PATTERNS, AND NEXT STEPS

ABSTRACT. For every real quadratic field $K = \mathbb{Q}(\sqrt{D})$, $D \le 300$ squarefree, whose Hermite–Humbert constant $\gamma_{K,2}$ (Icaza's multiplicative normalisation) has been certified in this project — 157 of the 182 fields — we list all classes of perfect binary Humbert forms modulo $\mathrm{GL}_2(\mathcal{O}_K)$ and scaling, and decide eutaxy for each. There are 55444 perfect classes, of which 15313 are extreme, 80 weakly eutactic and 40051 not eutactic; in every field the largest γ of a perfect form equals the certified $\gamma_{K,2}$. We describe the enumeration (a branch-and-bound over the fundamental region with certified discarding rules, followed by a local Newton solve), prove that it is complete under one explicitly stated numerical hypothesis, and discuss how to remove that hypothesis. The data suggest three regularities: the number of perfect classes is proportional to $\zeta_K(-1)$, i.e. to the volume of the Hilbert modular surface, with no regulator factor (in contrast with the additive normalisation); $\gamma_{K,2}$ is described to within about 20% (residual standard deviation 0.18 in log) by $d_K^{1.17}\mathrm{Reg}_K^{-0.61}$, and for $h_K = 1$ Thunder's lower bound reads $\gamma_{K,2} \ge 4\pi^2\zeta_K(-1)/(\sqrt{d_K}\,\mathrm{Reg}_K)$, for which we give an elementary proof; for $h_K > 1$ it is not known to bound $\gamma_{K,2}$ (it bounds a constant that dominates the projective one), but the data satisfy it even without its factor $1/h_K$; and forms with an automorphism of order 3 have $\frac{3}{4}\gamma$ equal to a norm from $\mathbb{Z}[\zeta_3]$, reflecting a CM structure. Every claim is labelled [PROVED], [CERTIFIED] (certified or exhaustive computation, range stated) or [CONJECTURE].

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1. Introduction

Let $K = \mathbb{Q}(\sqrt{D})$ be a real quadratic field with discriminant $d = d_K$, ring of integers $\mathcal{O}_K$, fundamental unit $\varepsilon_0 > 1$, regulator $\mathrm{Reg}_K = \log \varepsilon_0$ and class number $h = h_K$. A *binary Humbert form* is a pair $S = (S_1, S_2)$ of positive definite real 2×2 matrices, S_j attached to the real embedding $x \mapsto x^{(j)}$. Following Icaza and [1, 21, 22] we use the *multiplicative* normalisation (1)

$$S[v] = S_1[v^{(1)}]\, S_2[v^{(2)}], \quad m(S) = \min_{0 \neq v \in \mathcal{O}_K^2} S[v], \quad \gamma(S) = \frac{m(S)}{\sqrt{\det S_1 \det S_2}}, \quad \gamma_{K,2} = \sup_S \gamma(S).$$

This is a different theory from the *additive* normalisation $S_1[v^{(1)}] + S_2[v^{(2)}]$ of Koecher, Leibak and Gunnells–Yasaki [18, 19, 17]: the minimum is a product rather than a sum, so it is not a linear function of S, and there is no Ryshkov polyhedron. The two are never mixed below; the additive theory is the subject of the companion project `pav10/perfect_forms`, which we only quote for comparison. (In the additive theory Ong [14] classified the binary perfect forms over $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$ with Koecher's method: 2, 3, 2 classes, the same numbers as in the certified census of `perfect_forms`; Ash [2] uses the same Voronoi–Koecher decompositions of self-adjoint cones. Neither concerns the multiplicative forms studied here.)

Known values. Cohn [7] and Claus [6] computed the lowest points of the Hilbert modular fundamental domain for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$, which give $\gamma_{K,2}$ there (Claus determines them exactly, together with the boundary manifolds of the fundamental domain), and these values were established rigorously in [1]. Pohst and Wagner obtained $\mathbb{Q}(\sqrt{13})$ and the extreme forms of $\mathbb{Q}(\sqrt{17})$ having a unimodular pair of minimal vectors, without deciding whether these are all [21]; then $\mathbb{Q}(\sqrt{6})$ ($\gamma_{K,2} = 5$) and $\mathbb{Q}(\sqrt{21})$ ($\gamma_{K,2} = 16/3$), and conjectured the values for $\mathbb{Q}(\sqrt{7})$ and $\mathbb{Q}(\sqrt{11})$ [22]. Coulangeon, Icaza and O'Ryan computed $\gamma_{K,2}$ for the complex cubic field $\mathbb{Q}(\rho)$, $\rho^3 - \rho^2 + 1 = 0$ [9], and note that every value known then was obtained by constructing the possible sets of minimal vectors of perfect forms. In this project it was certified for 157 fields with $D \leq 300$ (branch-and-bound over the fundamental region, every certificate re-checked with outward rounding; see the project README). The present notes report a second computation on top of the first: the complete list of *perfect* forms (in Coulangeon's sense [16]) and their *eutaxy*, for all 157 fields. The extreme forms (perfect and eutactic) are the local maxima of γ; the global maximum is $\gamma_{K,2}$.

Summary.

(1) *Algorithm* (§3). Voronoi's neighbour algorithm is not available in the multiplicative setting (§3.1). We enumerate instead: a branch-and-bound over the region of highest points discards boxes with three certified rules, and the remaining small boxes are solved locally. Theorem 3.4 proves completeness under a single hypothesis (H) on the local Newton solve, and §3.5 explains how interval Newton would remove it. The cost is proportional to $\zeta_K(-1)$ (§3.4).
(2) *Results* (§4). 55444 perfect classes in 157 fields (Table 1); validation against the literature, against the certified $\gamma_{K,2}$, and by an independent re-run at half the box size.
(3) *Patterns* (§5). A volume law $n_{\mathrm{perf}} \approx 7.9\, \zeta_K(-1)$ (Conjecture 5.2); an empirical law and a proved lower bound for $\gamma_{K,2}$ (Proposition 5.4, after Thunder); CM structure of the symmetric forms (Observation 5.9).
(4) *Next steps* (§6) and the literature we need (§7).

Conventions. Labels: [PROVED] — proved here or in a cited source; [CERTIFIED] — established by a certified or exhaustive computation in the stated range (for the enumeration of perfect forms this means: complete under hypothesis (H) of §3.3); [CONJECTURE] — conjecture or heuristic. Fits are least-squares fits of logarithms; R^2 is the coefficient of determination. All numbers are produced by `analysis/patterns.py` from `results/perfect/` and `results/gamma_K2.tsv`.

2. Setting

2.1. **Forms as points of $\mathbb{H}^2$.** Up to the two independent scalings $(S_1, S_2) \mapsto (\lambda_1 S_1, \lambda_2 S_2)$, which leave γ unchanged, a form is a point $z = (z_1, z_2) \in \mathbb{H}^2$, $z_j = x_j + iy_j$, via $S_j \propto \frac{1}{y_j}\begin{pmatrix} 1 & x_j \\ x_j & |z_j|^2 \end{pmatrix}$. For $v = (a, b) \in \mathcal{O}_K^2$,

$$f_v(z) = \frac{|a^{(1)} + b^{(1)} z_1|^2 \, |a^{(2)} + b^{(2)} z_2|^2}{y_1 y_2}, \qquad G(z) = \min_{v \neq 0} f_v(z) = \gamma(S).$$

G is invariant under $\mathrm{GL}_2(\mathcal{O}_K)$ and under the Galois swap $(z_1, z_2) \mapsto (z_2, z_1)$, and $f_{\varepsilon v} = f_v$ for units ε, so minimal vectors come in *unit classes*. The space of forms modulo scaling is thus $\mathbb{H}^2$ itself (real dimension 4), and forms modulo equivalence are points of the Hilbert modular surface $\mathrm{PGL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$. (In the additive normalisation only the common scaling is free, and the space has dimension 5; see §5.1.)

2.2. **Highest points and the search region.** For $h = 1$ every orbit has a *highest* point (maximal $y_1 y_2$), and there $e_1 = (1, 0)$ is a minimal vector, so $G(z) = f_{e_1}(z) = 1/(y_1 y_2)$ [1, Prop. 1.1]. Using translations by $\mathcal{O}_K$, $z \mapsto \varepsilon_0^k z$, $x \mapsto -x$ (all in $\mathrm{PGL}_2(\mathcal{O}_K)$) and the Galois swap, every orbit has a highest point in

$$\mathcal{R} = \{x = r_1 + r_2\omega,\ r_1 \in [-\tfrac{1}{2}, \tfrac{1}{2}],\ r_2 \in [0, \tfrac{1}{2}];\ y_2/y_1 \in [1, \varepsilon_0];\ y_1 y_2 \geq 2/d\},$$

$\omega = (1 + \sqrt{D})/2$ or $\sqrt{D}$; the last condition is Cohn's bound $\gamma_{K,2} \leq d/2$ [7, 8]. For $h > 1$ the project uses one *chart* per pair of ideal classes $\{C, C^{-1}\}$: in the chart of an integral ideal $\mathfrak{c}$ of minimal norm in C the lattice is $\mathfrak{c}^{-1} \oplus \mathfrak{c}$, with the same region up to the translation lattice (project README, "Steinitz charts"). We write $t = \frac{1}{2}\log(y_1 y_2)$, $s = \frac{1}{2}\log(y_2/y_1)$, and use box coordinates (x_1, x_2, s, t); within a chart put $N_v = f_v/f_{e_1}$, so that $N_v(z) = |a^{(1)} + b^{(1)} z_1|^2 |a^{(2)} + b^{(2)} z_2|^2$.

2.3. **Perfection and eutaxy.** For a form S and a minimal vector v let

$$X_v = \big(S_2[v^{(2)}]\, v^{(1)} v^{(1)T},\ S_1[v^{(1)}]\, v^{(2)} v^{(2)T}\big),$$

the gradient of $S \mapsto S[v]$ in the 6-dimensional space $\mathrm{Sym}_2(\mathbb{R})^2$; $X_{\varepsilon v} = X_v$ for units. S is *perfect* if the X_v (v minimal) span a space of dimension 5, the maximum possible [16, 1]; *eutactic* if $(S_1^{-1}, S_2^{-1}) = \sum_v \rho_v X_v$ with all $\rho_v > 0$; *weakly eutactic* if this holds with $\rho_v \geq 0$ but not with $\rho_v > 0$; and *extreme* (a strict local maximum of γ) iff perfect and eutactic [16].

3. The enumeration

3.1. **Why not Voronoi's algorithm.** In the additive theory $S \mapsto S_1[v^{(1)}] + S_2[v^{(2)}]$ is linear, the forms with minimum ≥ 1 form a locally finite convex polyhedron (Ryshkov), perfect forms are its vertices, and Voronoi's algorithm walks from vertex to neighbouring vertex [23, 18, 17]. In the multiplicative theory the constraint $S[v] \geq 1$ is not linear (it is $\log S_1[v^{(1)}] + \log S_2[v^{(2)}] \geq 0$), the region $\{m \geq 1\}$ is not convex, and the local structure at a perfect point — although described by Coulangeon's theory — does not give a finite combinatorial "neighbour" step that is guaranteed to succeed. We implemented such a walk (`voronoi.py`: Lipschitz-certified steps along the edges of the arrangement $\{f_v = G\}$); it works on generic vertices but fails on degenerate faces, where more than the expected number of f_v meet. We therefore enumerate.

3.2. **The algorithm.** Fix a chart. Let $\mathcal{V}$ be a finite set of unit classes containing every class v with $f_v \le d/2$ somewhere on $\mathcal{R}$ (it exists and is computed exactly as in the $\gamma_{K,2}$-certificate). For a box B of (x_1, x_2, s, t)-space with $t \le 0$:

Rule 3: discard B if it misses the translation parallelogram;
Rule 1: discard B if $\sup_B N_v < 1$ for some $v \in \mathcal{V}$ (then e_1 is not minimal anywhere in B, so $B \cap \mathcal{R}' = \emptyset$, $\mathcal{R}'$ the set of points of $\mathcal{R}$ at which e_1 is minimal);
Rule 5: discard B if fewer than 5 classes $v \in \mathcal{V}$ (including e_1, for which $N_{e_1} \equiv 1$) satisfy $\inf_B N_v \le 1$.

The bounds $\sup_B N_v$, $\inf_B N_v$ are the product bounds of the $\gamma_{K,2}$-certificate (exact interval bounds for $|a + bz|^2$ on a box, evaluated in floating point with a relative safety margin 10^{-9}). Boxes that survive are bisected along the coordinate of largest *hyperbolic* width ($\Delta x_j/y_j$, Δs, Δt) until that width is $\le w_{\min} = 0.05$. In a surviving leaf B, with candidate set $C_B = \{v \ne e_1 : \inf_B N_v \le 1\}$:

(1) for every 4-subset $\{v_1, \dots, v_4\} \subset C_B$, run damped Newton on $\log N_{v_i}(q) = 0$, $i = 1, \dots, 4$, in $q = (x_1, x_2, \log y_1, \log y_2)$, from the centre of B and three random points of B; accept the first limit that lies in B (with a margin of 1% of the box size);
(2) polish an accepted point to 50 digits; compute the full set M of minimal classes *independently of* $\mathcal{V}$ by the exact short-vector enumeration; reject the point if e_1 is not minimal; compute the rank of $\{X_v\}_{v \in M}$ and keep the point if it is 5;
(3) decide eutaxy by a linear programme (and, when $|M| = 5$, by solving for the unique ρ at 50 digits).

Finally the perfect points from all charts are grouped into $\mathrm{GL}_2(\mathcal{O}_K)$-classes by [21, Alg. 11] in exact arithmetic, and the stabiliser of each class in $\mathrm{GL}_2(\mathcal{O}_K)/\mathcal{O}_K^\times$ is computed. Large fundamental units are handled by cutting the s-range into slabs.

3.3. **Correctness.**

Lemma 3.1 ([PROVED])**.** *Let $z \in \mathcal{R}'$ be perfect. Then z has a minimal class $v = (a, b)$ with $b \ne 0$, and $y_1 y_2 \le 1$, i.e. $t \le 0$; in the chart of $\mathfrak{c}$, $y_1 y_2 \le N(\mathfrak{c})^{-1} \le 1$. For $h = 1$ every minimal class other than that of e_1 has $b \ne 0$.*

Proof. For a minimal $v = (a, 0)$ we have $S_j[v^{(j)}] = a^{(j)2} S_j[e_1]$ and $S[v] = S[e_1] = m$, so $X_v = \left(\frac{m}{S_1[e_1]} E_{11}, \frac{m}{S_2[e_1]} E_{11}\right) = X_{e_1}$ ($E_{11} = e_1 e_1^T$). So classes with $b = 0$ add nothing to the rank, and a perfect point (rank 5) has a minimal class with $b \ne 0$. For it, $f_v \ge (b^{(1)} y_1)^2 (b^{(2)} y_2)^2 / (y_1 y_2) = N(b)^2 y_1 y_2$, while $f_v = f_{e_1} = 1/(y_1 y_2)$; hence $(y_1 y_2)^2 \le N(b)^{-2}$. For $h = 1$, $|N(b)| \ge 1$; in a chart $b \in \mathfrak{c}$, so $|N(b)| \ge N(\mathfrak{c}) \ge 1$. Finally, for $h = 1$ and $b = 0$, $f_v = N(a)^2 f_{e_1}$ forces $|N(a)| = 1$, so a is a unit and v is in the class of e_1. □

Lemma 3.2 ([PROVED])**.** *At $z \in \mathcal{R}'$ with minimal classes $M \ni e_1$, the rank of $\{X_v\}_{v \in M}$ is 5 if and only if the differentials $d \log N_v$, $v \in M \setminus \{e_1\}$, span the 4-dimensional cotangent space of $\mathbb{H}^2$ at z. In that case some 4 of them are linearly independent, and z is an isolated, nondegenerate solution of the corresponding 4×4 system $\log N_{v_i} = 0$.*

Proof. For minimal v, $\langle X_v, (S_1, 0)\rangle = \langle X_v, (0, S_2)\rangle = S[v] = m$, so all X_v lie in an affine subspace A of codimension 2 not containing 0. Hence $\operatorname{rank}\{X_v\} = 1 + \operatorname{rank}\{X_v - X_{e_1}\}$, and the differences lie in the annihilator of the two scaling directions, i.e. in the cotangent space of the quotient $\mathbb{H}^2$, of dimension 4. There $X_v - X_{e_1}$ is m times $d \log(S[v]/S[e_1]) = d \log N_v$. So rank 5 $\iff$ the $d \log N_v$ span a space of dimension 4, and the Jacobian of a suitable 4-subset is invertible at z. □

Lemma 3.3 ([PROVED], up to floating point)**.** *Rules 1, 3 and 5 never discard a box containing a perfect point $z \in \mathcal{R}'$ with $t \le 0$. Every minimal class of such a z lies in $\mathcal{V}$ and in the candidate set of every leaf containing z.*

Proof. At z, $N_v(z) \ge 1$ for all v (e_1 minimal), so $\sup_B N_v \ge 1$: rule 1 does not apply; z lies in the parallelogram: rule 3 does not apply; z has at least 5 minimal classes (perfection needs

rank 5), each with $N_v(z) = 1$, hence $\inf_B N_v \le 1$: rule 5 does not apply. A minimal class has $f_v(z) = G(z) \le \gamma_{K,2} \le d/2$, so it is in $\mathcal{V}$. The bounds are exact interval bounds evaluated in floating point with a safety margin; "up to floating point" refers to this evaluation, which is the same as in the $\gamma_{K,2}$-certificate (there re-checked with outward rounding). □

Hypothesis (H). *For every perfect point z and every leaf $B \ni z$, Newton for at least one 4-subset of minimal classes of z with invertible Jacobian (Lemma 3.2) converges to z from one of the four starting points, and no other solution in B of the same subset is reached first.*

Theorem 3.4 (completeness, [PROVED] under (H))**.** *Under (H) the enumeration returns a representative of every $\mathrm{GL}_2(\mathcal{O}_K)$-class of perfect binary Humbert forms, and only perfect forms.*

Proof. Every class has a highest point in some chart's region with e_1 minimal and, by Lemma 3.1, $t \le 0$. The leaves and discarded boxes tile the box cover of the region, and by Lemma 3.3 z lies in a leaf B whose candidate set contains its minimal classes. By (H) some 4-subset produces z; the polished point has the minimal set and rank computed independently of $\mathcal{V}$, so it is kept, and only points with rank 5 are kept. The classification is exact. □

Remark 3.5 (what is not rigorous). (i) Hypothesis (H): Newton has a nonsingular solution in the box but convergence from the chosen starts is not proved; the solve is local (box width ≤ 0.05 in hyperbolic terms) and started from 4 points, and a failure would need all of them to fail. (ii) The rank 5 and the eutaxy decisions are made at 50 digits with tolerances (10^{-25} relative for singular values, 10^{-9} for the eutaxy margin), so a weakly eutactic class is weakly eutactic to within 10^{-9}. (iii) The class list is checked against the certified $\gamma_{K,2}$ only through its maximum.

3.4. **Cost.** Over the 157 fields the search used $7.2 \cdot 10^8$ boxes, $5.3 \cdot 10^7$ leaves and $1.5 \cdot 10^9$ Newton runs. The number of boxes grows like $\zeta_K(-1)^{1.02}$ ($R^2 = 0.91$, the 127 fields with $d \ge 100$), the same law as the number of perfect classes (§5.1): the work is proportional to the volume of the Hilbert modular surface, i.e. roughly to $d^{3/2}$. The single most expensive field took 3.2 hours on one core ($D = 263$). Each class is found 3.1 times on average (different charts, translates and slabs), and 7.2% of the classes only once.

3.5. **Making it robust and faster.**

(1) *Certified completeness.* Replace (H) by an interval Newton (Krawczyk) test per 4-subset and leaf: either it proves that the subset has no solution in B, or that it has exactly one; leaves where neither holds are bisected. At a perfect point the Jacobian of some subset is invertible (Lemma 3.2), so near it the test succeeds after finitely many bisections; degenerate (non-perfect) crossings are rejected by the rank test. Together with an interval check of the rank and of the eutaxy LP this makes the lists certified.
(2) *Fewer subsets.* Most of the time goes to 4-subsets with no common zero in the leaf. Pairwise and triple tests (interval bounds on $\log N_v - \log N_w$) discard most of them before Newton.
(3) *Fewer copies.* Restricting the region by the stabilisers of the cusps (and the chart symmetries) would remove most of the 3.1 copies per class.
(4) *Parallelism inside a field.* Boxes are independent; splitting the s- or t-range over processes removes the long tail (now one process per field).
(5) *A compiled kernel* for the box bounds and the batched Newton. (The batched Newton already gave a factor ≈ 3 over the per-subset loop.)
(6) *A multiplicative neighbour walk* with exact handling of degenerate faces would make the cost proportional to the output; a mass formula over the cells (as in the additive census of `perfect_forms`) would then give an independent completeness check.

3.6. **Is the algorithm new?** The ingredients are known: highest points and the floor of the fundamental domain [7, 1, 22], perfection and eutaxy in this setting [16], the equivalence test [21, Alg. 11], and the idea of solving $S[v] = 1$ for a few candidate minimal vectors. The earlier algorithms differ from ours as follows.

- [21, Alg. 9] and [1, 9] normalise $e_1, e_2 \in M(S)$ (a *unimodular pair* of minimal vectors), enumerate globally all 3-sets of candidates from a finite set M determined by norm bounds [21, Lemmas 7, 8], solve the polynomial systems $S[u_i] = 1$ (three unknowns b_1, b_2, c), and keep the solutions that are perfect and eutactic. This finds the *extreme* forms with a unimodular pair; that every extreme form has one must be proved field by field ([21, Lemma 16] for $\mathbb{Q}(\sqrt{13})$; [9, Props. 2.9, 2.10] via Lenstra's constant), and for $\mathbb{Q}(\sqrt{17})$ it was left open.
- [22, §2] steps through Cohn's description of the fundamental domain with a prescribed increment, collects the minimal vectors met in a set M, and solves the systems for all 4-sets of M; there is no criterion that M contains all the vectors needed [22, p. 940].
- For imaginary quadratic K (binary hermitian forms, $\mathbb{H}^3$ in place of $\mathbb{H}^2$), [5] compute the projective constant and its extreme forms for $|d_K| < 70$ from a cellular fundamental domain of the Bianchi group (Mendoza, Vogtmann), as the lowest points of that domain. This is the closest analogue we know of a geometric computation over the whole fundamental domain; it determines γ and the forms attaining it, not all perfect forms.
- Here only e_1 is normalised (it is minimal at the highest point, for any class number, chart by chart), so no unimodular-pair hypothesis is needed. For $h_K > 1$ minimal vectors need not be unimodular (Bergé's example over $\mathbb{Q}(\sqrt{10})$, [3, §2]), but the content ideal of a minimal vector has least norm in its class [3, §2], which is why the charts are indexed by ideals of least norm; the 4-subsets are formed *locally*, in boxes of hyperbolic width ≤ 0.05 that survive certified discarding rules, so that the candidate set provably contains the minimal classes of every perfect point in the box (Lemma 3.3); and all perfect forms are kept, not only the eutactic ones.

To our knowledge this combination, and complete lists of multiplicative perfect forms for more than a few fields, are new. We have not seen Wagner's tables [22] (cited there as a web page) or later work by the same authors.

4. Results

Table 1. Totals over the 157 fields with certified $\gamma_{K,2}$, $D \leq 300$ [certified].

perfect classes	55444
extreme (perfect + eutactic)	15313
weakly eutactic (in 52 fields)	80
not eutactic	40051
fields with one perfect class	$D = 2, 3, 5$
most perfect classes	$D = 274$: 1467
median fraction extreme	0.332

4.1. **Validation.**

- [certified] In every one of the 157 fields the largest γ of a perfect class equals the certified $\gamma_{K,2}$ (relative difference $\leq 10^{-12}$).
- [certified] Agreement with the literature. $D = 2, 3, 5$: one class each [1] ($|\mathrm{Aut}| = 10$ for $D = 5$), with $\gamma = (12 + 8\sqrt{6})/15$, 4, $4/\sqrt{5}$, the reciprocals of the heights $N(y) = (2\sqrt{6}-3)/4$, $1/4$, $\sqrt{5}/4$ of Claus's lowest points [6, Sätze 6, 3, 5]. $D = 13$: two perfect classes, both extreme, $\gamma = (84 + 6\sqrt{91})/35 = \sqrt{(1476 + 144\sqrt{91})/175}$ (critical) and $\gamma = 4.00045\ldots$ (five minimal classes), the forms in rows 2 and 1 of the table of [21, p. 916]; the form of row 3 has perfection rank < 5 there and is correctly absent. $D = 17$: exactly the two extreme forms of [21, pp. 918–919], $\gamma = \sqrt{(784 + 128\sqrt{34})/225} = 2.6079893\ldots$ and $(40 + 8\sqrt{85})/45 = \sqrt{(1408 + 128\sqrt{85})/405} = 2.5279190\ldots$, and no other perfect class; so the question left open there is settled and $\gamma_{\mathbb{Q}(\sqrt{17}),2} = 2.6079893\ldots$ (the $\gamma_{K,2}$-certificate

confirms the value). $D = 6$, 21: $\gamma_{K,2} = 5$, 16/3 as in [22]; $D = 7$, 11: the conjectured values $\sqrt{(5584 + 640\sqrt{70})/605} = (100 + 16\sqrt{70})/55$ and 28/3 of [22, p. 946] are the certified ones. $D = 10$: 7 perfect, 5 extreme (test suite). Where the experimental neighbour walk completed, it found the same classes.

- [CERTIFIED] Independent re-run: the fields with $D \le 77$ listed in the file `robustness.txt` of `results/perfect/` were recomputed with half the leaf width ($w_{\min} = 0.025$, hence different leaves and Newton starts); the class lists (value of γ to 25 digits, number of minimal classes, eutaxy, |Aut|) agree in every field.
- An audit after the short-vector fix (project history): one record changed ($D = 46$ had a spurious duplicate class with one minimal vector missed; now 84 perfect, 39 extreme).

4.2. **The table.** Table 2 in the appendix lists n_{perf} and n_{ext} for every field; the file `D⟨D⟩.json` in `results/perfect/` contains every class (value of γ, closed form when PSLQ finds one, minimal vectors, rank, eutactic coefficients, |Aut|, a point).

5. Patterns

5.1. A volume law for the number of perfect forms.

Observation 5.1 ([CERTIFIED], 157 fields). Over the 127 fields with $d \ge 100$ the log-log slope of n_{perf} against $\zeta_K(-1)$ is 1.017 ($R^2 = 0.984$); against $\zeta_K(-1)\mathrm{Reg}_K$ it is 0.807 ($R^2 = 0.892$), against d it is 1.525 ($R^2 = 0.973$), against $h\mathrm{Reg}_K$ 1.604 ($R^2 = 0.54$). For the 98 fields with $d > 200$, $n_{\text{perf}}/\zeta_K(-1)$ has median 7.90, quartiles 7.37, 8.56 and range $[5.96, 10.86]$; the mass $\sum 1/|\mathrm{Aut}|$ behaves the same way (median 7.74). For the extreme forms the slope against $\zeta_K(-1)$ is 0.823.

Conjecture 5.2 ([CONJECTURE]). *$n_{\text{perf}}(K) \asymp \zeta_K(-1)$ as $d_K \to \infty$, with constants independent of Reg_K and h_K.*

Heuristic. By Siegel's theorem as stated in [11, §1.3, (9)] the Euler volume (Gauss–Bonnet integral) of $\mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$ is $2\zeta_K(-1)$, and its hyperbolic volume (measure $\prod_j dx_j\, dy_j/y_j^2$) is $2\pi^{-2}d_K^{3/2}\zeta_K(2) = 8\pi^2\zeta_K(-1)$ (*loc. cit.*; the formula for $\zeta_K(-1)$ used in `analysis/patterns.py` is [11, §1.4, (11), (12)]); $\zeta_K(-1) \asymp d^{3/2}L(2, \chi_d)$. In the multiplicative normalisation the space of forms modulo scaling is exactly $\mathbb{H}^2$, so a locally finite configuration of perfect points with bounded local density has $\asymp$ vol classes. In the additive normalisation the separate scalings of S_1 and S_2 are not symmetries; the space has one more dimension, the log-unit direction, and `perfect_forms` (FINDINGS §3, additive, $d \le 100$) finds $n \asymp \zeta_K(-1)\mathrm{Reg}_K$ there; for unary forms Yasaki's bound [24] relates the count to the period of a continued fraction, i.e. to Reg_K. The three laws are consistent with "number of perfect forms $\asymp$ volume of the space of forms modulo $\mathrm{GL}_n(\mathcal{O}_K)$ and the symmetries of the normalisation".

The constant 7.9 is not explained. An Euler-characteristic argument would need the full cell structure (edges and higher cells), which the multiplicative theory does not provide (§3.1); a neighbour walk with cells would turn the heuristic into a mass formula and a completeness certificate.

5.2. The Hermite–Humbert constant.

Observation 5.3 ([CERTIFIED], 157 fields). A least-squares fit gives

$$\log \gamma_{K,2} \approx 1.172 \log d - 0.614 \log \mathrm{Reg}_K - 1.711 \qquad (R^2 = 0.977,\ \text{residual s.d. } 0.18),$$

against $R^2 = 0.923$ for $\log d$ alone; adding $\log h$ changes nothing (coefficient -0.05). $\gamma_{K,2}/d$ ranges from 0.068 ($D = 46$) to 0.358 ($D = 5$), well below Cohn's $\frac{1}{2}$. The low values of $\gamma_{K,2}/d$ for $D \equiv 1 \pmod 8$ (2 split) are explained by the regulator: the residuals of the fit have mean -0.04 (2 split), -0.03 (2 inert), $+0.02$ (2 ramified).

Known bounds. Upper: $\gamma_{K,2} < d/2$ (Cohn, as quoted in [21, 22]; see [7, 8]; the non-strict form is the case $m = n = 2$ of $\gamma_{K,n} \le m^{-m}\gamma_{mn}^{m}|d_K|$ [4, Thm. 4.1, Rem. 4.2], with $\gamma_4^2 = 2$) and $\gamma_{K,2} \le 16d/\pi^2$ ([12, Thm. 1], $n = 2$, $r = 2$, $s = 0$). Lower: $\gamma_{K,2} \ge B(K)$, the least norm of a nontrivial integral ideal, whenever Lenstra's constant satisfies $L(K) < 4$ [9, Prop. 4.1]; for quadratic K, $B(K) \le 4$. For $h_K > 1$: $\gamma_{K,2} \ge N(I)N(J)$ for ideals I, J of least norm in mutually inverse non-principal classes (proof of [4, Cor. 3.2], stated there as $\gamma_{K,2} \ge 2\mathcal{N}(K)$); since I^σ lies in the inverse class, $N(J) = N(I)$, so $\gamma_{K,2} \ge \mathcal{N}(K)^2$ with $\mathcal{N}(K)$ the largest least norm of an ideal class. A lower bound linear in $\sqrt{d_K}$ is attributed to a preprint of Baeza in [12, p. 21] (not seen). The strongest is the Minkowski–Hlawka-type bound of Thunder [15, Thm. 2]: with $n = 2$, $d = 1$ (his notation), $r = 2$, $s = 0$, $w = 2$ it reads $\gamma_2(K)^2 \ge \zeta_K(2)\, d_K/(\pi^2 h_K \mathrm{Reg}_K)$. How $\gamma_2(K)^{[K:\mathbb{Q}]}$ compares with $\gamma_{K,2}$ is settled in [5, §5]: with the projective constant $\gamma_K^{\mathrm{p}} = \sup_S \min_v S[v]/(N((v))^2\sqrt{\det S})$, where (v) is the ideal generated by the coordinates of v, one has $\gamma_K^{\mathrm{p}} \le \gamma_{K,2}$ and $\gamma_K^{\mathrm{p}} \le \gamma_2(K)^{[K:\mathbb{Q}]}$, with equality in both when $h_K = 1$. (Thunder's height divides by the norm of the content and his maximum runs over all h_K components of $\mathrm{GL}_2(\mathbb{A}_K)$, i.e. also over non-free lattices.) So for $h_K = 1$, with $\zeta_K(-1) = d^{3/2}\zeta_K(2)/(4\pi^4)$,

$$\gamma_{K,2} \;\ge\; \frac{\zeta_K(2)\, d_K}{\pi^2 h_K \mathrm{Reg}_K} \;=\; \frac{4\pi^2\zeta_K(-1)}{h_K\sqrt{d_K}\,\mathrm{Reg}_K}, \tag{2}$$

while for $h_K > 1$ (2) is proved for $\gamma_2(K)^2$ but not for $\gamma_{K,2}$; we keep the factor h_K in the formula to compare the data with Thunder's bound. For $h_K = 1$ the following elementary argument gives the same bound; it explains why the regulator enters: the cusp has length Reg_K, so a fixed volume spread over a longer cusp forces the floor of the fundamental domain (height $1/\gamma_{K,2}$) up.

Proposition 5.4 ([PROVED] for $h_K = 1$, given Siegel's volume formula; the case $h_K = 1$ of [15])**.** $\gamma_{K,2} \ge 4\pi^2\zeta_K(-1)/(\sqrt{d_K}\,\mathrm{Reg}_K)$.

Proof. Every orbit has a highest point $z \in \mathcal{R}$, and there $G(z) = 1/(y_1y_2) \le \gamma_{K,2}$, so the set $\mathcal{R}(Y) = \{z \in \mathcal{R} :\ y_1y_2 \ge Y\}$ with $Y = 1/\gamma_{K,2}$ meets every orbit of the group Γ generated by $\mathrm{PGL}_2(\mathcal{O}_K)$ and the Galois swap. Hence its hyperbolic volume is at least that of $\Gamma\backslash\mathbb{H}^2$. The x-part of $\mathcal{R}$ has area $\sqrt{d}/2$ (half a period parallelogram of $\mathcal{O}_K$, of area $\sqrt{d}$). With $p = \log(y_1y_2)$, $q = \log(y_2/y_1)$, $dy_1dy_2/(y_1y_2)^2 = \frac{1}{2}e^{-p}\,dp\,dq$ and $q \in [0, \mathrm{Reg}_K]$, so $\mathrm{vol}\,\mathcal{R}(Y) = \frac{\sqrt{d}}{2}\cdot\frac{\mathrm{Reg}_K}{2}\cdot\frac{1}{Y} = \frac{\sqrt{d}\,\mathrm{Reg}_K\,\gamma_{K,2}}{4}$. On the other hand $[\mathrm{PGL}_2(\mathcal{O}_K) : \mathrm{PSL}_2(\mathcal{O}_K)] = |\mathcal{O}_K^\times/\mathcal{O}_K^{\times 2}| = 4$ and the swap has index 2, so $\mathrm{vol}(\Gamma\backslash\mathbb{H}^2) = 8\pi^2\zeta_K(-1)/8 = \pi^2\zeta_K(-1)$ (Siegel's formula: the hyperbolic volume of $\mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$ is $2\pi^{-2}d_K^{3/2}\zeta_K(2) = 8\pi^2\zeta_K(-1)$, [11, §1.3]). □

Remark 5.5 (imaginary quadratic fields)*.* The same covering argument works for binary hermitian forms over an imaginary quadratic field K with $h_K = 1$, in the normalisation of [5]. A form corresponds to a point (x, r) of $\mathbb{H}^3$ with $S[e_1]/\sqrt{\det S} = 1/r$ at a highest point; the highest points lie over a fundamental domain of Γ_∞ in $\mathbb{C}$, of area $\sqrt{|d_K|}/(2w')$ with $w' = |\mathcal{O}_K^\times|/2$, and $\int_{1/\gamma}^{\infty} dr/r^3 = \gamma^2/2$. With Humbert's formula $\mathrm{vol}(\mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3) = |d_K|^{3/2}\zeta_K(2)/(4\pi^2)$, whose proof was completed by Grunewald and Kühnlein [10], this gives $\gamma_K^2 \ge w_K\zeta_K(2)|d_K|/(2\pi^2)$, which is Thunder's bound [15, Thm. 2] with $r = 0$, $s = 1$, $h_K = 1$. Against the exact values for the nine fields with $h_K = 1$ [5, Table 1] the ratio γ_K/bound grows from 1.13 ($d_K = -3$) to 2.13 ($d_K = -163$) [CERTIFIED], much as in Observation 5.6 for real quadratic fields.

Observation 5.6 ([CERTIFIED], 157 fields)*.* All these bounds hold in the data. $\gamma_{K,2}$ divided by (2) lies in $[1.46, 14.5]$ (minimum at $D = 5$, then $D = 2, 17$); for $h_K > 1$, where (2) is not a theorem for $\gamma_{K,2}$, the ratio is ≥ 4.35, and even without the factor $1/h_K$ it is ≥ 2.17. $\gamma_{K,2}/B(K) \ge 1.053$ for $D \ne 5$ (nearly attained at $D = 2$). $\gamma_{K,2}/(d/2) \le 0.72$ and $\gamma_{K,2}/(16d/\pi^2) \le 0.22$. For the 87 fields with $h_K > 1$, $\gamma_{K,2}/\mathcal{N}(K)^2 \ge 2.18$ (minimum at $D = 10$; $\mathcal{N}(K)$ ranges up to 10), so the bound of [4] is within a factor of about 2 of the truth across the range, much closer than

(2) for large d. The ratio to (2) grows with d (log-log slope of $\gamma_{K,2}$ against $\zeta_K(-1)/(\sqrt{d}\mathrm{Reg}_K)$: 1.27).

Remark 5.7. Since $\zeta_K(2) \in [\zeta(4), \zeta(2)^2]$, (2) gives $\gamma_{K,2} \gg d/(h\mathrm{Reg}_K)$, and $h\mathrm{Reg}_K \ll \sqrt{d}\log d$ by the class number formula, so $\gamma_{K,2} \gg \sqrt{d}/\log d$ for real quadratic K with $h_K = 1$; with $\gamma_{K,2} < d/2$ this brackets the growth. The data sit near $d^{1.17}\mathrm{Reg}_K^{-0.61}$ and do not depend on h. Questions: is $\gamma_{K,2} \asymp d/\mathrm{Reg}_K$ (the fit says no)? Does (2) hold for $\gamma_{K,2}$ when $h_K > 1$, and can its factor $1/h_K$ be removed (the data say yes to both, with room to spare)? Is $\limsup \gamma_{K,2}/d < \frac{1}{2}$?

5.3. **Minimal vectors and automorphisms.**

Observation 5.8 ([CERTIFIED]). Of the 55444 perfect classes, 44643 (80.5%) have exactly 5 minimal classes, the minimum for perfection (Coulangeon's bound $\#M(S) \geq rn(n+1)/2 + sn^2 - (r+s-1) = 5$, [9, (2–1)], [21, Cor. 4]); 8472 have 6; the maximum is 24 ($D = 263$). The stabiliser in $\mathrm{GL}_2(\mathcal{O}_K)/\mathcal{O}_K^\times$ is trivial for 53228 classes, of order 2 for 2106, and of order $3, 4, 6, 10, 24$ for $4, 64, 40, 1, 1$ classes.

Observation 5.9 ([CERTIFIED], 157 fields). All 45 classes whose stabiliser has order divisible by 3 have $\frac{3}{4}\gamma$ equal to an integer of the form $a^2 + ab + b^2$ (a norm from $\mathbb{Z}[\zeta_3]$). Of the 64 classes with stabiliser of order 4, 51 have γ an integer of the form $a^2 + b^2$. The stabiliser of order 10 is that of the unique class of $\mathbb{Q}(\sqrt{5})$ ($\gamma = 4/\sqrt{5}$), and the one of order 24 that of $\mathbb{Q}(\sqrt{3})$ ($\gamma = 4$).

Proposition 5.10 ([PROVED]). *Let $g \in \mathrm{GL}_2(\mathcal{O}_K)$ stabilise a form and have order 3 in $\mathrm{PGL}_2(K)$. Then $g^2 - tg + \delta = 0$ with $t \in \mathcal{O}_K$, $\delta = \det g = t^2$, and $\mathcal{O}_K^2$ is a module over the order $\mathcal{O}_K[g] \cong \mathcal{O}_K[\frac{1}{2}t(1+\sqrt{-3})]$ of the CM field $L = K(\sqrt{-3})$.*

Proof. g^3 is scalar and g is not, so the eigenvalues are $\lambda, \lambda\zeta_3$; then $t = \lambda(1+\zeta_3) = -\lambda\zeta_3^2$ and $\delta = \lambda^2\zeta_3$, so $t^2 = \delta$ and the discriminant is $t^2 - 4\delta = -3t^2$; $t \neq 0$ since δ is a unit, so $g = \frac{t}{2}(1+\sqrt{-3})$ in $L = K(\sqrt{-3})$. □

Heuristically, a form fixed by g is, in each embedding, invariant under a rotation of order 3, hence a multiple of the hexagonal form, and $S[v]$ becomes a norm form of L on a rank-one $\mathcal{O}_K[g]$-module. Since $L = K(\sqrt{-3}) \supset \mathbb{Q}(\sqrt{-3})$, norms from L to $\mathbb{Q}$ factor through $\mathbb{Q}(\zeta_3)$; we expect this to explain the shape $\frac{4}{3}(a^2+ab+b^2)$ in general [CONJECTURE] (note $\frac{4}{3} = \gamma_2^2$, the square of the Hermite constant of the hexagonal lattice). The same mechanism with ζ_4, ζ_5 ($\sqrt{5} \in K$) and ζ_{12} ($\sqrt{3} \in K$) accounts for the other symmetric forms. This is the correspondence with quadratic (CM) extensions of K asked about: forms with extra automorphisms carry the structure of hermitian forms over CM extensions L/K (Proposition 5.10 for order 3), and we expect their γ to be governed by norms from L [CONJECTURE]. Binary hermitian forms over an imaginary quadratic field are the case $K = \mathbb{Q}$ of this picture; their (projective) Hermite constants are tabulated in [5, Table 1], and the relative version over L/K is what a proof of the conjecture would need.

Elliptic points. Since forms modulo scaling are the points of $\mathbb{H}^2$, a perfect class with $|\mathrm{Aut}| > 1$ is a perfect point fixed by an elliptic element. For g as in Proposition 5.10, t is a unit ($t^2 = \det g$), so $g/t \in \mathrm{SL}_2(\mathcal{O}_K)$ has trace 1 and order 3 in $\mathrm{PSL}_2(\mathcal{O}_K)$: every class with $3 \mid |\mathrm{Aut}|$ is an elliptic point of order 3 of the Hilbert modular group. Ong asked, in the additive setting, whether the elliptic fixed points correspond to perfect forms [14]. Here the answer is mostly no [CERTIFIED]: for the 71 fields with $(D, 6) = 1$, $D \geq 7$, Prestel's formula ($a_3 = h(-3D)$ or $h(-12D)$ for $D \equiv 1$ or $3 \bmod 4$, as quoted in [11, p. 198]) gives 822 classes of elliptic points of order 3 in total, and only 22 perfect classes have $3 \mid |\mathrm{Aut}|$ (at most $4 \cdot 22 = 88$ of the 822 classes, since a $\mathrm{GL}_2(\mathcal{O}_K)$-orbit splits into at most $|\mathcal{O}_K^\times/\mathcal{O}_K^{\times 2}| = 4$ orbits of $\mathrm{PSL}_2(\mathcal{O}_K)$). For $\mathbb{Q}(\sqrt{3})$ the lowest point is an elliptic fixed point of order 12 [6, Satz 3] (stabiliser of order 24 in $\mathrm{GL}_2(\mathcal{O}_K)/\mathcal{O}_K^\times$ here), but lowest points need not be elliptic: for $\mathbb{Q}(\sqrt{2})$ Claus notes that his two lowest points modulo the Hilbert modular group are not elliptic fixed points [6, §4].

5.4. **Eutaxy.** The median fraction of extreme classes per field is 0.33, against 0.97 for perfect unary forms over real quadratic fields (`perfect_forms`, FINDINGS §5); weakly eutactic classes

occur (80, in 52 fields), unlike the unary case. The number of extreme classes grows more slowly than n_{perf} (slope 0.82 against $\zeta_K(-1)$).

6. Next steps

6.1. **Completing $D \le 300$ and certifying the lists.** The 25 fields with $D \le 300$ whose $\gamma_{K,2}$ is not certified (large units) are the obvious gap; the enumeration itself would run on them (it does not need $\gamma_{K,2}$, only Cohn's bound), but its cost grows with the length of the cusp. Certified completeness needs the interval-Newton version of §3.5 (estimated cost: a small constant factor, since most leaves have no solution and are rejected after one interval evaluation).

6.2. **Ternary forms over real quadratic fields.** The space of ternary forms modulo the two scalings is $(\mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3)^2$, of dimension 10. A box enumeration in dimension 10 is not feasible: the number of leaves scales like $w_{\min}^{-10}$ times the volume of $\mathrm{PGL}_3(\mathcal{O}_K)\backslash(\mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3)^2$, which already for $\mathbb{Q}(\sqrt{5})$ is far beyond the binary case. Feasible routes: (a) a multiplicative Voronoi walk (§3.5, item 6), whose cost is proportional to the output; (b) a search for $\gamma_{K,3}$ alone, by local maximisation from many seeds together with structured candidates (the lattices whose trace form is similar to E_6, or hermitian structures over CM extensions as in Proposition 5.10), with a branch-and-bound certificate only if a good candidate is found; (c) in the additive normalisation the Voronoi algorithm works in any rank, and `perfect_forms` is the place for it. We would start with (b) for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$. A free upper bound comes from the Mordell inequality of Baeza and Icaza [3, §3]: $\gamma_{K,n}^{n-2} \le \gamma_{K,n-1}^{n-1} M_{n,K}^{2(n-1)/n}$, where $M_{n,K}$ bounds the norm of the content of minimal vectors ($M_{n,K} = 1$ for $h_K = 1$). For $h_K = 1$ this gives $\gamma_{K,3} \le \gamma_{K,2}^2$, so the 70 fields of class number one in our table come with a certified upper bound for $\gamma_{K,3}$ at no cost (e.g. $\le \gamma^2_{\mathbb{Q}(\sqrt{5}),2}$), against which a candidate from (b) can be measured; and $\gamma_{K,4}^2 \le \gamma_{K,3}^3$ links the next step.

6.3. **Binary forms over totally real fields of higher degree.** For a totally real cubic field the space of binary forms modulo the three scalings is $\mathbb{H}^3$ (dimension 6), $\gamma_{K,2}$ is again the reciprocal height of the floor, and every step of §3 generalises (products of three factors, ε in a rank-2 unit lattice, 6-subsets for Newton). The cost of the $\gamma_{K,2}$-certificate and of the enumeration is proportional to the volume ($\asymp \zeta_K(-1)$ again) times $w_{\min}^{-6}$ instead of $w_{\min}^{-4}$: feasible for the smallest cubic fields ($d = 49, 81, 148, 169, \dots$), out of reach for large ones without the improvements above. Fields with complex places are also natural first targets: a complex place contributes a factor $\mathbb{H}^3$, and [9] gives a value to test against ($\mathbb{Q}(\rho)$, $\rho^3 - \rho^2 + 1 = 0$, $d_K = -23$: $\gamma_{K,2} \approx 2.4684958820$) and two further fields ($x^3 + x^2 - x + 1$, $d_K = -44$; $x^4 + x^3 + x^2 + x + 1$, $d_K = 125$) where the same method applies. The code is written for two embeddings throughout; a degree-n version is a rewrite of `humbert.py`, `certify.py` and `perfect.py`, not a patch.

6.4. **Further questions suggested by the data.**

(1) Prove $n_{\mathrm{perf}} \asymp \zeta_K(-1)$ (Conjecture 5.2), perhaps first a lower bound by a covering argument as in Proposition 5.4.
(2) Explain $\gamma_{K,2} \approx 0.18\, d^{1.17}\mathrm{Reg}_K^{-0.61}$; prove (2) for $\gamma_{K,2}$ when $h_K > 1$, and without the factor $1/h_K$.
(3) Classify the symmetric forms by CM extensions L/K (which L, which orders, which ideal classes) and compute their γ from norms in L.
(4) The set $W \subset \mathbb{H}^2$ of points with at least two minimal classes has dimension 3, the virtual cohomological dimension of $\mathrm{PSL}_2(\mathcal{O}_K)$ (Borel–Serre, as recalled in [2, §1]); the perfect points are its 0-dimensional strata (Lemma 3.2). Is W a $\mathrm{GL}_2(\mathcal{O}_K)$-equivariant deformation retract of $\mathbb{H}^2$, as Ash's retract is for the linear (Koecher) cones [2]? If so, the perfect classes found here are the vertices of a finite cell complex for the cohomology of the Hilbert modular group.

(5) Compare the multiplicative and additive extreme forms field by field: `perfect_forms` finds $\gamma^4_{\mathrm{add}}/(16\gamma^2_{\mathrm{mult}}) \in [1.56, 46.1]$, never 1.

7. Literature

Read for this version (provided by the project owner): [21, 22, 9, 12, 15, 3, 4, 5, 11, 6, 14, 2, 10]. They fix the novelty statement (§3.6), the validation (§4.1), the known bounds, the attribution of the lower bound to Thunder and its restriction to $h_K = 1$ (§5.2, via [5, §5]), the role of non-unimodular minimal vectors for $h_K > 1$ and the Mordell inequality for the ternary step [3], the lower bound for $h_K > 1$ and the source of the $d/2$ bound [4], the volume, $\zeta_K(-1)$ and elliptic-point formulas [11], the exact lowest points for $D = 2, 3, 5$ [6], the imaginary quadratic volume formula [10], and the additive results kept apart from ours [14, 2].
Still wanted (cited above only as reported by others):

(1) R. Baeza, *The volume of the space of Humbert reduced forms*, preprint, Universidad de Chile, 1994 (cited in [12] for a lower bound linear in $\sqrt{d_K}$ and in [3] for a lower bound by zeta values) — the closest relative of Proposition 5.4.
(2) S. Ohno and T. Watanabe [13] (bounds for $\gamma_{K,n}$, used in [9]).
(3) Bounds for $M_{3,K}$: [4] estimates only $M_{K,2}$ ($4\mathcal{N}(K)^2 \le M_{K,2}$), and the ternary Mordell bound above needs $M_{3,K}$ for $h_K > 1$.
(4) M. Wagner's thesis and the table cited in [22]; any later enumeration of multiplicative perfect forms.
(5) G. van der Geer's book on Hilbert modular surfaces (cusps, elliptic points, and the orbifold structure needed to turn the volume heuristic of §5.1 into a proof); Siegel's original papers as cited in [11].
(6) Perfect and extreme hermitian forms over CM fields (for Proposition 5.10), e.g. the generalisations of Voronoi's theorem [20]; T. Watanabe's constants of algebraic groups and the Mendoza and Vogtmann cell structures, as cited in [5].

The additive theory [18, 19, 17] is in the companion project.

Appendix A. Counts per field

Table 2: Per field: D, h_K, $\gamma_{K,2}$ (rounded), number of perfect and of extreme classes [CERTIFIED].

D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext
2	1	2.106	1	1	87	2	76	210	52	187	2	105.4	810	207
3	1	4	1	1	89	1	10.45	28	10	190	2	100.8	619	171
5	1	1.789	1	1	91	2	33.98	210	75	194	2	146.1	721	228
6	1	5	3	3	93	1	20.27	17	7	195	4	190.9	658	139
7	1	4.252	4	4	95	2	101.3	241	72	197	1	38.56	67	31
10	2	8.706	7	5	97	1	9.204	31	19	202	2	119.1	832	164
11	1	9.333	8	7	101	1	20.43	31	12	203	2	157	671	170
13	1	4.035	2	2	102	2	65.78	266	87	205	2	48.08	88	30
14	1	11.11	11	6	103	1	43.52	378	119	206	1	149.1	886	220
15	2	12.55	13	11	105	2	25.6	36	10	209	1	17.33	95	38
17	1	2.608	2	2	106	2	49.65	303	107	210	4	161.4	714	226
19	1	8.056	16	9	107	1	89.33	291	99	213	1	44.95	72	39
21	1	5.333	2	2	109	1	9.542	26	17	215	2	193.3	819	183
22	1	10.8	23	16	110	2	90.91	238	70	218	2	129.9	729	184
23	1	17.33	25	16	111	2	64.51	340	113	219	4	196	899	208
26	2	23.55	36	18	113	1	12.07	43	23	221	2	57.78	108	52
29	1	5.792	3	3	114	2	82.67	358	97	222	2	161.4	822	220
30	2	28	38	17	115	2	101.3	303	69	223	3	148.7	906	245
31	1	11.31	37	17	118	1	50.13	431	120	226	8	207.7	976	280
33	1	4.571	3	3	119	2	100.8	373	111	227	1	169.3	801	248

Table 2: (continued)

D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext
34	2	34.01	78	38	122	2	129.4	301	60	229	3	50.7	95	33
35	2	33.12	41	15	123	2	81.73	368	104	230	2	241.3	844	160
37	1	5.736	4	4	129	1	12.31	51	21	231	4	218.7	1032	263
38	1	30.16	34	19	130	4	117.6	439	114	233	1	21.55	143	54
39	2	25	54	31	131	1	91.24	465	114	235	6	281.3	1037	250
41	1	5.321	10	5	133	1	25.35	49	33	237	1	52.05	98	36
42	2	36.63	70	27	134	1	91.66	524	135	238	2	137.5	1357	280
43	1	21.13	87	27	137	1	13.29	47	22	246	2	90.05	1091	363
46	1	12.5	84	39	138	2	103.4	365	93	247	2	106.6	1181	240
47	1	33.41	73	38	141	1	28.09	63	27	253	1	29.8	129	43
51	2	36.6	91	44	142	3	106.2	448	137	254	3	184	1075	322
53	1	12.42	9	1	143	2	138.4	339	76	255	4	244.1	1058	257
55	2	40.11	118	48	145	4	35.14	102	36	257	3	52.21	145	47
57	1	9.333	10	5	146	2	129.3	520	165	258	2	204.3	1074	262
58	2	40.1	123	34	149	1	26.8	50	14	263	1	179.5	1144	303
59	1	44.1	110	45	154	2	78.99	534	153	265	2	34.11	246	78
61	1	6.456	8	7	155	2	125.8	531	118	266	2	146.7	1012	250
62	1	52.08	111	47	157	1	20.82	62	26	267	2	133.2	1124	320
65	2	17.07	21	7	158	1	109.1	578	142	269	1	49.51	126	43
66	2	52	142	47	159	2	111.2	565	164	273	2	33.33	267	113
67	1	29.27	177	65	161	1	12.63	79	32	274	4	209.6	1467	273
69	1	13.16	15	7	165	2	41.72	52	21	277	1	34.4	151	67
70	2	52.23	133	31	167	1	129.3	477	152	278	1	201	1095	296
71	1	50.08	210	67	170	4	158.8	566	170	282	2	183.6	1157	276
73	1	4.977	17	15	173	1	35.02	48	29	285	2	66.22	122	61
74	2	47.85	143	37	174	2	124	523	176	287	2	199.4	1134	358
77	1	17.63	12	8	177	1	14.3	88	40	290	4	279.4	1125	289
78	2	57.48	162	50	178	2	110	760	217	291	4	208.6	1427	398
79	3	52.14	203	74	181	1	17.92	68	35	293	1	70.92	109	39
82	4	75.75	228	57	182	2	176.7	539	111	298	2	165.3	1431	349
83	1	62.3	181	64	183	2	109	611	177	299	2	198.5	1223	304
85	2	22.1	47	14	185	2	36.13	106	16					
86	1	58.86	271	84	186	2	101	647	139					

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